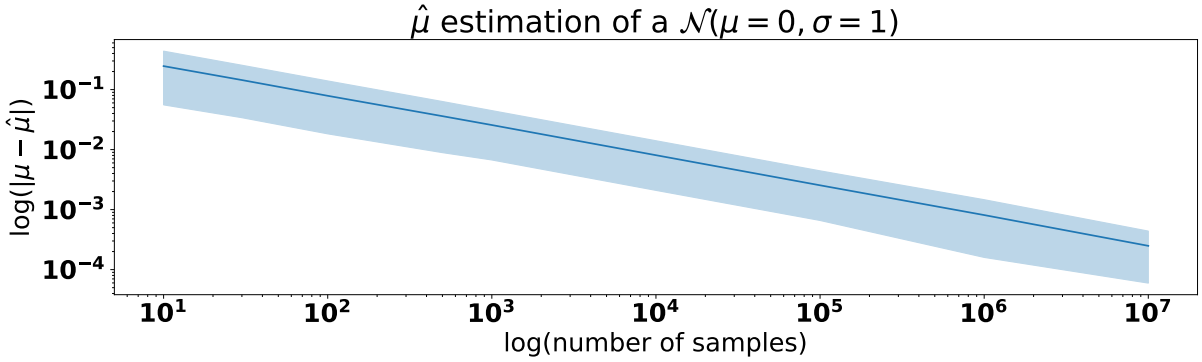


Figure 12: Approximation of a single-dimensional Gaussian $\mathcal{N}(0, 1)$ as a function of number of points. The mean estimator and its standard deviation are calculated from running the procedure 10000 times.

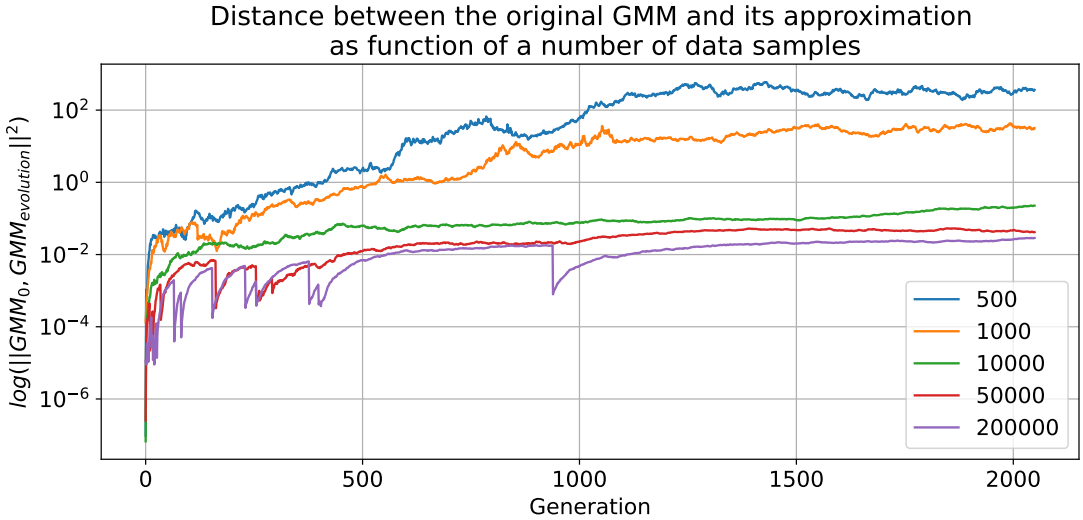


Figure 13: Progressive fitting of a GMM with different number of samples. On the y-axis is shown the logarithm of $L2$ distance between the two GMM distributions. Over the generations the distance begins to grow and can become quite large. The jumps in the distance for large sample sizes occur due to the fixed number of iterations and precision for the expectation maximization algorithm.

Assumptions:

3*. The extra noise is going to be assumed to be bounded and of the order larger than the sample mean deviation. To be precise we will have a constant K (not dependent on generation i), such that for all i:

$$\|\varepsilon_{i+1}\| \leq \frac{K}{M_i} \tag{18}$$

Now with the alternative assumption in place, we can follow the exact same calculations to arrive at

$$\mathbb{E}\left[R_{W_2}^{i+1}\right] \geq \mathbb{E}\left(\|\mu_i - \mu\|^2\right) + \frac{\operatorname{Tr}\Sigma}{M_i} + \mathbb{E}\left(\|\varepsilon_{i+1}\|^2\right) + \frac{2}{\sqrt{M_i}}\mathbb{E}\left((\varepsilon_{i+1})^\top \Sigma_i^{1/2} T^{i+1}\right) \tag{19}$$

Similar to before, we need to evaluate (which we instead bound this time):

$$\frac{2}{\sqrt{M_i}}\mathbb{E}\left((\varepsilon_{i+1})^\top \Sigma_i^{1/2} T^{i+1}\right) = \frac{2}{\sqrt{M_i}} \int d\Sigma_i \; p(\Sigma_i) \operatorname{Tr}\left[\Sigma_i^{1/2} \operatorname{Cov}(\varepsilon_{i+1}, T^{i+1}|\Sigma_i)\right] \neq 0 \tag{20}$$

$$\geq -\frac{2\sqrt{N}}{\sqrt{M_i}} \int d\Sigma_i \; p(\Sigma_i) \sqrt{\operatorname{Tr}\left[\Sigma_i \Sigma_{\epsilon_{i+1}}\right]} \tag{21}$$

$$\geq -\frac{2\sqrt{N}}{\sqrt{M_i}} \sqrt{\mathbb{E}\left(\varepsilon_{i+1}^\top \Sigma_i \varepsilon_{i+1}\right)}, \tag{22}$$

$$\geq -\frac{2\sqrt{N}}{\sqrt{M_i}} \sqrt{\frac{K^2 \operatorname{Tr}\Sigma}{M_i^2}} = \frac{-2K\sqrt{N}}{M_i\sqrt{M_i}} \sqrt{\operatorname{Tr}\Sigma}, \tag{23}$$

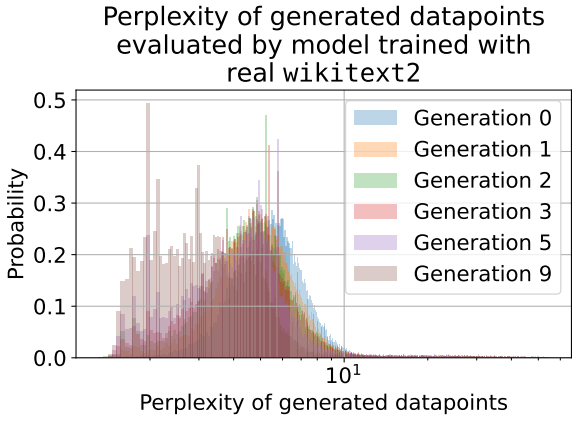
where we used the Cauchy-Schwarz and Jensen inequalities. Note that this is far from optimal inequality, since instead of using the expected value of the largest eigenvalue, we instead bounded it by $\operatorname{Tr}\Sigma$. In particular, the per step bound is then:

$$\mathbb{E}\left[R_{W_2}^{i+1}\right] \geq \mathbb{E}\left(\|\mu_i - \mu\|^2\right) + \frac{\operatorname{Tr}\Sigma}{M_i} + \mathbb{E}\left(\|\varepsilon_{i+1}\|^2\right) - \frac{2K\sqrt{N}}{M_i\sqrt{M_i}} \sqrt{\operatorname{Tr}\Sigma}. \tag{24}$$

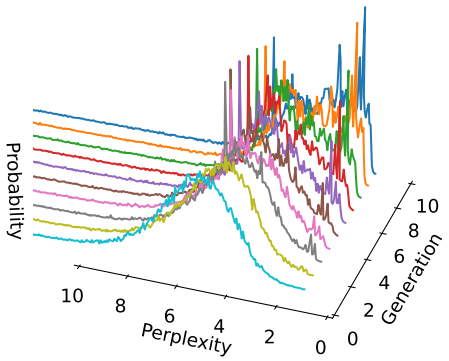
Without knowledge of the specific values of K, N or $\operatorname{Tr}\Sigma$, the best we can do is consider what this means for the bound as M_i becomes large. In particular, contribution from the last two terms will be of order at most $3/2$. As a result we recover a bound similar to all of the ones observed so far:

$$\mathbb{E}_{\mu_{n+1},\sigma_{n+1}^2}\left[R_{W_2}\right] \geq \operatorname{Tr}\Sigma \left(\frac{1}{M_0} + \frac{1}{M_1} + \cdots + \frac{1}{M_n}\right) + \mathcal{O}(3/2) \tag{25}$$

In particular, we find in the same way, that superlinear scaling would be required to minimise the lower bound on *model collapse* even in the case of more generic models of approximation, in which the mean at step $i+1$ can be separated into the sample mean and an extra bounded term of order at most $1/M_i$.

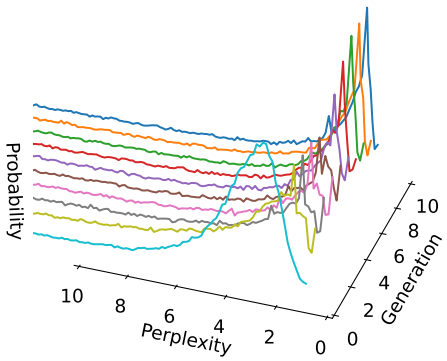


(a) Overlaid histograms

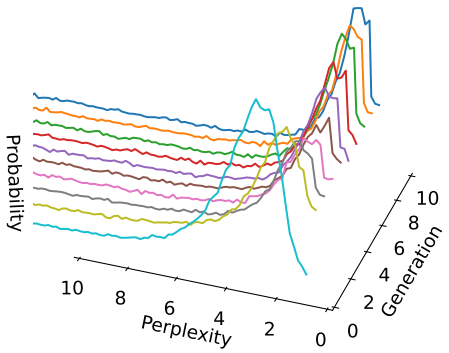


(b) 3D view

Figure 14: Histogram of perplexities of each individual data training sequence produced by different generations as is evaluated by the very first model trained with the real data. Over the generations models tend to produce samples that the original model (trained with real data) is more likely to produce. At the same time, a much longer tail appears for later generations – later generations start producing samples that would never be produced by the original model *i.e.* they start misperceiving reality based on errors introduced by their ancestors. Models here are explicitly forced to not repeat sequences with a penalty of 2.0.



(a) Figure 11.a in 3D. No data preserved.



(b) Figure 11.b in 3D. 10% original data preserved.

Figure 15: Histogram of perplexities of each individual data training sequence produced by different generations as is evaluated by the very first model trained with the real data. Over the generations models tend to produce samples that the original model (trained with real data) is more likely to produce. At the same time, a much longer tail appears for later generations – later generations start producing samples that would never be produced by the original model *i.e.* they start misperceiving reality based on errors introduced by their ancestors.